



DPUDerivative

Computes the derivative of an input value numerically.

1. Inputs:

In Input (function)

2. Outputs:

Out Output (derivative)

3. Functional principle:

Let $Out[i]$ be the value of *Out* in the i th simulation step. Then

$$Out[i] = \frac{In[i] - In[i - 1]}{Period}$$

Wherein *Period* is the length of one simulation step in seconds.

4. Example:

You can derive the speed from the distance driven between two simulation steps:

```
DPUDerivative CalcSpeed
{
    Computer = { VDYN };
    Index = 10;
};
Connections = {
    VDyn.s -> CalcSpeed.In
};
```